CSE752: Deep Neural Networks

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Introduction to Neural Network

Computation of a Single Neuron: $f(w^\top x + b)$

A neuron computes $f(w^\top x + b)$, because the input is usually a vector.

- **Single input (scalar):** $a = f(wx + b)$
- **Multiple inputs (vector $x \in \mathbb{R}^d$):** $z = w^\top x + b$, then $a = f(z)$

The neuron does two things: an affine step (weighted sum plus bias) that gives the pre-activation z, then a nonlinearity f applied to that scalar. The nonlinearity is the only place a curve can enter. If f were the identity, stacking layers would collapse into one affine map.

For a whole layer, the same idea is written with a matrix: $z = Wa^{(l-1)} + b$, $a = f(z)$, with f applied elementwise.

Number of Inputs to a Neuron

A neuron can take multiple values as input. A single input is just the special case $d = 1$.

With d inputs, the neuron has **one weight per input** but only **one bias**: $x \in \mathbb{R}^d$, $w \in \mathbb{R}^d$, $b \in \mathbb{R}$. It multiplies each input by its own weight, sums the results, adds the bias, and applies f:

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d + b = w^\top x + b, \qquad a = f(z)$$

Whatever the number of inputs, the output is always **one scalar**.

For example, in a $784 \to 128 \to 64 \to 10$ network, each neuron in the first hidden layer takes all 784 inputs, so it has 784 weights and 1 bias. This is why a layer's parameter count is $n_{\text{out}} \cdot n_{\text{in}} + n_{\text{out}}$: n_{out} neurons, each with n_{in} weights and one bias.

What Is a Layer?

A layer is a group of neurons that all receive the same input vector and compute in parallel, each producing one scalar. Stacking those scalars gives the layer's output vector, which becomes the input to the next layer.

If a layer has n_{out} neurons and receives n_{in} inputs, then every neuron has its own n_{in} weights and its own bias. Stacking the weight vectors as rows gives one matrix, so the whole layer computes in a single step:

$$z = Wa_{\text{prev}} + b, \qquad a = f(z)$$

with W of shape $(n_{\text{out}} \times n_{\text{in}})$, and b, z, a of length n_{out}. The activation f is applied to each coordinate independently.

Example: a $2 \to 2 \to 1$ network.

- **Input layer:** just x, the raw inputs. It has no weights and does no computation.
- **Hidden layer:** 2 neurons, each seeing both inputs ($W^{(1)}$ is 2×2). It is called hidden because its outputs are neither the inputs nor the final prediction.
- **Output layer:** 1 neuron, seeing both hidden outputs ($W^{(2)}$ is 1×2), producing $\hat{y}$.

A layer where every neuron sees every input from the previous layer is called a **fully connected** (or dense) layer. A network with several of these in a row is an MLP.

Do Input Layer Neurons Perform Computation?

No. The input layer's "neurons" are just placeholders that hold the raw input values $(x_1, x_2, \dots)$. They have no weights, no bias and no activation, so nothing is computed there.

The first real computation happens in the first hidden layer, which takes the input values and applies $f(Wx+b)$. The weight matrix $W^{(1)}$ is described as connecting the input layer to the first hidden layer, but it belongs to the hidden layer's neurons: its shape is $(n_{\text{hidden}} \times n_{\text{input}})$.

This is also why a $2 \to 2 \to 1$ network is usually called a "2-layer network" (hidden + output). The input layer is drawn as circles for visualization, but it is not counted because it does no computation.

Input Layer Size for a Tabular Dataset

Suppose a tabular dataset has 1000 rows and 4 columns. The number of input neurons depends on what the columns are.

- **All 4 columns are input features (target is separate):** 4 input neurons, one per feature. The number of rows does not matter; each row is passed in as a vector $x \in \mathbb{R}^4$, and the 1000 rows are 1000 training examples.
- **One of the 4 columns is the target (label):** only 3 are features, so the input layer has 3 neurons. The target column is what the output layer is compared against in the loss, not something fed in.
- **Preprocessing can change the input size:** a categorical column with k categories, one-hot encoded, becomes k input neurons instead of 1. Dropping a column, or adding engineered features, changes the count too.

The rule is: **input layer size = number of numeric input values per row after preprocessing**, not the raw column count. The label is excluded.

Since there are 1000 rows, they are typically passed in batches (say 32 at a time). The input is then a matrix of shape $(\text{batch} \times n_{\text{features}})$, but the network's structure is unchanged: batch size never affects the number of neurons or weights.

Output Layer Design Based on the Label Type

For a dataset with 3 numerical features and 1 label, the input layer has 3 neurons. The label is not fed in; it only appears in the loss, where it is compared against the network's output.

The output layer depends on the label type:

- **Regression (continuous label):** 1 output neuron, usually with no activation (or identity).
- **Binary classification:** 1 output neuron with sigmoid.
- **Multi-class classification with k classes:** k output neurons with softmax.

A network such as $3 \to 8 \to 4 \to 1$ would work for regression or binary classification. The hidden sizes (8 and 4) are a free choice; only the input size (3) and the output size (set by the task) are fixed by the data.

One-Hot Encoding of Categorical Features

A neuron computes $w^\top x + b$, so every input has to be a number, and the number's size has to mean something. A categorical column like color $= \{\text{red}, \text{green}, \text{blue}\}$ is just labels. If we encode them as red $= 1$, green $= 2$, blue $= 3$, we tell the network that blue is "three times red" and green sits "between" them. The weight multiplies that number, so the network is forced to treat the categories as points on a line with a fixed order and spacing, which is usually false.

One-hot encoding avoids this by giving each category its own 0/1 column:

color	is_red	is_green	is_blue
red	1	0	0
green	0	1	0
blue	0	0	1

One original column becomes k columns (here $k = 3$), and exactly one of them is 1 in each row. Since the input layer has one neuron per input value, those k columns need k input neurons.

This also fixes the weight problem. The first hidden neuron now has a separate weight for each category: $w_{\text{red}}, w_{\text{green}}, w_{\text{blue}}$. Its pre-activation is

$$z = w_{\text{red}} \cdot \text{is_red} + w_{\text{green}} \cdot \text{is_green} + w_{\text{blue}} \cdot \text{is_blue} + (\text{other terms}) + b$$

For a red row, only w_{red} contributes. For a green row, only w_{green} does. So the network learns an independent effect for each category, with no imposed ordering or spacing.

Example: a dataset has 3 numerical features plus one categorical column with 4 categories. After one-hot encoding, each row is a vector of length $3 + 4 = 7$, so the input layer has 7 neurons.

Notes:

- If a column has thousands of categories (like a zip code), one-hot becomes very wide. In that case, **embeddings** are used, which map each category to a short learned vector instead.
- If the categories have a natural order (small $<$ medium $<$ large), encoding them as 1, 2, 3 can be reasonable, because the ordering is real.

Why Each Feature Gets One Input Slot Instead of Feeding All Features into One Neuron

The input layer is not made of computing neurons. It is just the input vector, with one slot per value, so "one neuron per feature" simply means "one slot per number." A neuron can take multiple inputs, but that happens in the next layer, not in the input layer.

Feeding all features into one neuron is possible, but that neuron is a hidden-layer neuron, and it has two consequences:

1. **It compresses the data.** With 3 features, one neuron computes $z = w_1x_1 + w_2x_2 + w_3x_3 + b$ and outputs a single scalar. Everything the next layer sees is that one number, so many different feature combinations collapse to the same value and information is lost. For example, with $w = (1, 1, 1)$, the rows $(1, 2, 3)$ and $(3, 2, 1)$ are indistinguishable.
2. **It limits what the network can learn.** The network can only use one weighted combination of the features. A layer with several neurons learns several different combinations (each with its own weight vector), and later layers can combine those into more complex patterns. One neuron gives only a single "view" of the data.

So the structure is:

- **Input layer:** holds the raw values, 3 slots, no computation, nothing lost.
- **First hidden layer:** every neuron takes all 3 features as input, and the number of neurons (a free choice) sets how many combinations are learned.

The input layer must keep the features separate so that the hidden neurons can each apply their own weights to each individual feature.

Weight Matrix Shapes: Example $4 \to 4 \to 3 \to 1$

Consider a dataset with 1000 rows and 5 columns (4 numeric features, 1 label), and a network with architecture $4 \to 4 \to 3 \to 1$.

Hidden layer 1. The weight matrix has shape (4×4) using the convention W of shape $(n_{\text{out}} \times n_{\text{in}})$. The layer has 4 neurons ($n_{\text{out}} = 4$) and receives the 4 input features ($n_{\text{in}} = 4$). Each row of W holds one neuron's 4 weights, and the bias b has length 4. The layer computes $z = Wx + b$, with W (4×4) times x (4×1) giving z (4×1).

Parameters for the whole architecture:

- Hidden layer 1: $W^{(1)}$ is 4×4, $b^{(1)}$ has length $4 \to 16 + 4 = 20$ parameters
- Hidden layer 2: $W^{(2)}$ is 3×4, $b^{(2)}$ has length $3 \to 12 + 3 = 15$ parameters
- Output layer: $W^{(3)}$ is 1×3, $b^{(3)}$ has length $1 \to 3 + 1 = 4$ parameters

Total: 39 parameters.

Weight Matrix Shape Convention: $(n_{\text{out}} \times n_{\text{in}})$ vs. $(n_{\text{in}} \times n_{\text{out}})$

Why is $W^{(2)}$ written as 3×4 and not 4×3? It depends on the convention. Here we use $(n_{\text{out}} \times n_{\text{in}})$, where each row holds one neuron's weights.

Hidden layer 2 has 3 neurons and receives 4 inputs:

- Each neuron needs 4 weights (one per input), so each neuron is a row of length 4.
- There are 3 neurons, so there are 3 rows.
- $W^{(2)}$ is therefore 3×4.

This makes the math work with the column-vector form $z = Wa + b$: (3×4) times (4×1) gives (3×1), a vector with one value per neuron.

The alternative 4×3 is the other common convention, $(n_{\text{in}} \times n_{\text{out}})$, where each *column* holds one neuron's weights. It is valid, but the computation is then written as $z = a^\top W + b$, i.e. (1×4) times (4×3) gives (1×3). PyTorch's `nn.Linear` stores the weight as $(n_{\text{out}} \times n_{\text{in}})$, while Keras's `Dense` stores it as $(n_{\text{in}} \times n_{\text{out}})$. The two are transposes of each other, with the same 12 numbers.

Whichever convention is chosen, it must be used consistently so that the shapes multiply correctly. In general, a layer's weight shape is (neurons in this layer $\times$ neurons in the previous layer) under the first convention.

Neurons and Logistic Regression

The big picture: **one neuron = a weighted score + a squashing step**. The affine score is introduced first, logistic regression shows it is a one-neuron network, two worked examples use real numbers, and finally we derive why one formula turns out so clean.

The Neuron as an Affine Score

Plain idea: A neuron takes some numbers in, mixes them using "importance" values (weights), adds a nudge (bias), then passes the result through a bending function.

Think of grading a student: score = $(2 \times \text{exam}) + (1 \times \text{homework}) + 5$ bonus points. That is the first half. The second half might be "cap the final grade between 0 and 1."

Equation: Neuron Score and Activation

$$z = w^\top x + b, \qquad a = f(z)$$

Symbols:

- x = the inputs, a list of d numbers (e.g., exam mark, homework mark)

- w = the weights, a list of d numbers, one per input. Bigger weight means that input matters more
- $w^\top x$ = multiply each input by its weight, then add everything up
- b = bias, a single number added at the end (a shift)
- z = the raw score (called the pre-activation)
- f = the activation function, a rule that reshapes z
- a = the neuron's final output

What it says: First compute a straight-line score z. Then pass z through f to get the output a.

Conditions: Always holds. "Affine" just means "straight-line style": plotted against any single input, z is a straight line.

Special values:

- $w = 0$: the neuron ignores the input, and z is just b.
- $b = 0$: the line passes through the origin, so zero input gives zero score.
- f = identity ($f(z) = z$): the neuron stays purely straight-line, no curves at all.

Equation: Stacking Affine Layers Collapses

$$W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$$

Symbols:

- W_1, W_2 = weight tables (matrices) for layer 1 and layer 2
- b_1, b_2 = the biases of each layer
- x = the input

What it says: If you stack two straight-line layers with no bending function between them, you could replace both with **one** straight-line layer. The left side is "do layer 1, then layer 2." The right side is the same thing as a single layer with weights W_2W_1 and bias $W_2b_1 + b_2$. (You get it by multiplying out the brackets.)

Why it matters: Without a bending function between layers, a 100-layer network is no more powerful than a 1-layer one. The activation function f is what makes depth useful.

Conditions: Holds whenever the matrix sizes fit together so the multiplication is possible.

Special values: If $W_1 = 0$, everything collapses to just b_2, so the input is ignored. If $W_2 = 0$, same result.

Logistic Regression as a One-Neuron Network

Plain idea: You want a yes/no answer (spam or not, $y = 1$ or 0). A neuron gives you any number, but you want a probability between 0 and 1. The sigmoid squeezes any number into that range. Then you need a way to measure how wrong the prediction was.

Equation: Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Symbols:

- σ (sigma) = the sigmoid function

- z = the raw score from the neuron
- e = a special constant ≈ 2.718
- e^{-z} = e raised to the power minus z

What it says: It takes any number and squashes it into the range between 0 and 1, which can be read as "probability of yes."

Conditions: Works for every real number z.

Special values:

- $z = 0$: $e^0 = 1$, so $\sigma = 1/2$. Total uncertainty.
- $z \to$ very large positive: $e^{-z} \to 0$, so $\sigma \to 1$ (confident "yes").
- $z \to$ very large negative: $e^{-z} \to$ huge, so $\sigma \to 0$ (confident "no").
- It never actually reaches 0 or 1, which is why the range is written as $(0, 1)$.

Equation: Prediction

$$\hat{y} = \sigma(w^\top x + b)$$

Symbols: $\hat{y}$ (y-hat) is the predicted probability. The rest are as before.

What it says: Compute the straight-line score, then squash it. That is the whole prediction.

Equation: Binary Cross-Entropy Loss

$$L = -\Big[\, y \log \hat{y} + (1 - y) \log(1 - \hat{y}) \,\Big]$$

This is the "how wrong was I?" number.

Symbols:

- L = loss (the penalty; lower is better)
- y = the true answer (0 or 1)
- $\hat{y}$ = your predicted probability
- log = natural logarithm

What it says: Because y is either 0 or 1, one half always disappears:

- If $y = 1$: $L = -\log \hat{y}$. Penalty is small when $\hat{y}$ is near 1, big when $\hat{y}$ is near 0.
- If $y = 0$: $L = -\log(1 - \hat{y})$. Penalty is small when $\hat{y}$ is near 0, big when $\hat{y}$ is near 1.

So it punishes you for being confident and wrong.

Conditions: Needs $y \in \{0, 1\}$ and $\hat{y}$ strictly between 0 and 1.

Special values:

- $\hat{y} = 1/2$: $L = \log 2 \approx 0.693$ either way (a coin-flip guess).
- $y = 1$ and $\hat{y} \to 0$: $\log \hat{y} \to -\infty$, so $L \to \infty$. Infinite penalty for being completely sure and wrong.
- $\hat{y}$ exactly 0 or 1: $\log(0)$ is undefined, which is why code "clips" $\hat{y}$ slightly away from the edges.

Two Key Facts (Previewed)

$$\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big), \qquad \frac{\partial L}{\partial z} = \hat{y} - y$$

Symbols:

- $\sigma'(z)$ = the slope of the sigmoid at z (how fast it changes)
- $\partial L/\partial z$ = how much the loss changes if you nudge z a tiny bit

What they say: The sigmoid's slope can be written using its own output. And the loss's sensitivity to z is just "prediction minus truth," which is very simple. The derivation below proves the second one.

Worked Example: Logistic Forward Pass

Setup: $w = 2$, $b = -2$, $x = 1$, $y = 1$ (true answer: yes).

Step by step:

- $z = wx + b = 2 \cdot 1 - 2 = \mathbf{0}$
- $\hat{y} = \sigma(0) = \mathbf{1/2}$
- $L = -\log(1/2) = \ln \mathbf{2} \approx \mathbf{0.693}$

(The loss formula reduces to $-\log \hat{y}$ because $y = 1$, so the second half vanishes.)

Reading the number: $\hat{y} = 1/2$ means the neuron is basically guessing, like flipping a coin. A loss of $\ln 2 \approx 0.693$ is the "coin flip loss." If your classes are balanced (half yes, half no) and you start with random weights, your starting loss should be near 0.693. If it starts at, say, 50, something is likely wrong.

Worked Example: Logistic Backward Pass

Plain idea: "Backward" means asking: *which way should I nudge w and b to reduce the loss?* The answer is the slope (gradient) of the loss with respect to each.

Step by step (same numbers: $\hat{y} = 1/2$, $y = 1$, $x = 1$):

1. $\dfrac{\partial L}{\partial z} = \hat{y} - y = \dfrac{1}{2} - 1 = -\dfrac{1}{2}$. The negative sign means: *increasing z would lower the loss.*
2. $\dfrac{\partial L}{\partial w} = \dfrac{\partial L}{\partial z} \cdot x = -\dfrac{1}{2} \cdot 1 = -\dfrac{1}{2}$. Since $z = wx + b$, a nudge to w changes z by x times as much, so you multiply by x. This is the chain rule: multiply the slopes along the path.
3. $\dfrac{\partial L}{\partial b} = \dfrac{\partial L}{\partial z} = -\dfrac{1}{2}$. A nudge to b changes z by exactly that amount (the factor is 1).

Equation: Gradient Descent Update

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$

Symbols:

- $\leftarrow$ = "replace with the new value"
- η (eta) = learning rate, the step size (a small positive number you choose)
- $\partial L/\partial w$ = the slope computed above

What it says: Move w a small step *opposite* to the slope. Here: $w \leftarrow 2 - \eta(-1/2) = \mathbf{2} + \eta/\mathbf{2}$.

Result: w grows, so z on this input becomes positive, so $\hat{y}$ moves up toward 1 (the correct answer).

Conditions: η must be positive. If it is too big you overshoot; too small and learning is slow.

Special values:

- $\eta = 0$: nothing changes, no learning.
- Gradient $= 0$: no movement, which happens at a minimum, a flat spot, or a "dead" path.

Derivation of $\partial L/\partial z = \hat{y} - y$

Plain idea: The loss depends on $\hat{y}$, and $\hat{y}$ depends on z. So the chain is $z \to \hat{y} \to L$. To find how L changes with z, find the slope of each link and multiply them (chain rule):

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z}$$

Setup

- $L = -\big[\, y \log \hat{y} + (1-y) \log(1-\hat{y}) \,\big]$ is the loss
- $\hat{y} = \sigma(z)$ is the prediction
- $\sigma(z) = \dfrac{1}{1+e^{-z}}$ is the sigmoid
- y is the true label (0 or 1), a fixed number, not something that changes with z

Step 1: Slope of the Sigmoid, $\partial \hat{y}/\partial z$

Write $\sigma(z) = (1+e^{-z})^{-1}$. Differentiate:

$$\sigma'(z) = (1+e^{-z})^{-2} \cdot e^{-z}$$

Split this into two fractions:

$$\sigma'(z) = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}$$

The first fraction is $\sigma(z) = \hat{y}$. For the second, note that $1 - \sigma(z) = \dfrac{e^{-z}}{1+e^{-z}}$, so it equals $1-\hat{y}$. Therefore:

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y})$$

Step 2: Slope of the Loss with Respect to $\hat{y}$, $\partial L/\partial \hat{y}$

Differentiate each term (the derivative of $\log u$ is $1/u$, and the derivative of $\log(1-\hat{y})$ picks up a minus sign):

$$\frac{\partial L}{\partial \hat{y}} = -\left[\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}\right] = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

Put both fractions over the common denominator $\hat{y}(1-\hat{y})$:

$$= \frac{-y(1-\hat{y}) + (1-y)\hat{y}}{\hat{y}(1-\hat{y})}$$

Expand the top: $-y + y\hat{y} + \hat{y} - y\hat{y} = \hat{y} - y$ (the $y\hat{y}$ terms cancel). So:

$$\frac{\partial L}{\partial \hat{y}} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$

Step 3: Multiply

$$\frac{\partial L}{\partial z} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})} \cdot \hat{y}(1-\hat{y})$$

The $\hat{y}(1-\hat{y})$ on top and bottom cancel:

$$\frac{\partial L}{\partial z} = \hat{y} - y$$

Sanity Check

With the worked example's numbers, $\hat{y} = 1/2$ and $y = 1$ give $\hat{y} - y = -1/2$, which matches the backward pass above.

Conditions

This holds only for **sigmoid output + binary cross-entropy** together, with $0 < \hat{y} < 1$ (otherwise Step 2 divides by zero). With a different loss or activation, the factors would not cancel.

Special Values

- $\hat{y} = y$: gradient is 0, so there is nothing to fix.
- $y = 1$, $\hat{y} \to 0$ (confident and wrong): gradient $\to -1$, a strong push. It does not vanish, even though $\sigma' \to 0$ and $\partial L / \partial \hat{y} \to \infty$ individually. The two extremes cancel exactly.

Why Squared Error Is a Poor Match for Sigmoid

If you used $L = \frac{1}{2}(\hat{y} - y)^2$ instead, the same chain gives:

$$\frac{\partial L}{\partial z} = (\hat{y} - y)\, \sigma'(z)$$

Nothing cancels, so there is a leftover $\sigma'(z)$.

Why that is a problem: When the neuron is very confident (z very large or very small), $\sigma'(z) \approx 0$. So even if the neuron is **confidently wrong**, the gradient is nearly zero and the weights barely move. Learning stalls exactly when it is needed most. Binary cross-entropy cancels that σ' factor, which is why sigmoid + BCE is the standard pairing for yes/no problems.

Summary

A neuron = straight-line score + squash; sigmoid turns scores into probabilities; cross-entropy punishes confident mistakes; and the gradient of that pair is simply *prediction − truth*, which tells you which direction to nudge the weights.

Convolutional Neural Networks (CNNs)

Problem 2: Locality Is Ignored

The core claim: an MLP is allowed to connect every unit to every pixel, but in images only nearby pixels matter for most useful features. The MLP has to discover this from data, whereas a CNN has it built in.

Sentence by sentence

1. **"The statistics of natural images are local."** Whether pixel $(10, 10)$ belongs to an edge depends on pixels around it, such as $(9, 10)$, $(11, 10)$ and $(10, 11)$. A pixel at $(30, 2)$ tells you almost nothing about it. Pixel values are strongly correlated with their neighbours, and the correlation fades quickly with distance.
2. **"Edges, corners and textures are properties of small neighbourhoods."** An edge is just a sharp intensity change between adjacent pixels, so a 3×3 or 5×5 patch is enough to detect it. Larger structures like faces or cats are built later by combining many of these local detections (the feature hierarchy in Section 10).

3. **"An MLP must learn from data that most of its weights should be nearly zero."** A hidden unit in a dense layer has one weight for every input pixel. For a $32 \times 32 \times 3$ CIFAR image, that is 3072 weights per unit. A unit that should detect a small edge only needs about 75 of them (a $5 \times 5 \times 3$ patch). The other $\sim$3000 should be zero, but nothing in the architecture says so. The network must push them toward zero through training.
4. **"Connecting each unit only to a small blob of pixels, instead of being told so in advance."** "Told in advance" means building the assumption into the architecture. That is what the notes call an inductive bias.

Why this is a real problem

- **Flattening hides the structure.** After flattening, the MLP sees 3072 numbers with no notion of which are neighbours. Two vertical neighbours end up 32 positions apart (Section 2.2). If you applied the same random shuffle to the pixels of every image, the MLP would learn equally well (or equally badly), because it never used the layout anyway. Locality is a fact about images that the MLP cannot see.
- **It is a learning problem, not a representation problem.** A dense layer can represent a local detector by setting the right weights to zero. This is exactly the matrix view in Section 6.4, where a convolution is a dense layer with a sparse, weight-tied matrix. So the MLP is not incapable of it. The difficulty is finding that solution from finite, noisy data.
- **The cost is overfitting and wasted capacity.** The weights that should be zero never become exactly zero. They pick up noise from the training set, which hurts generalisation, and they take up capacity that could detect something useful.

How CNNs fix it

Choice 1 (local connectivity) hard-codes the zeros: each output unit reads only a $k \times k$ patch, and the other links do not exist. In Example 2.1 this alone cuts the weights from $16{,}384 \times 3072 = 50{,}331{,}648$ to $16{,}384 \times 75 = 1{,}228{,}800$, a factor of about 41.

The three problems are separate. Problem 1 is about the sheer count of weights. Problem 2 (this one) is about the structure of those weights, which should be local. Problem 3 is about re-learning the same detector at every position, which Choice 2 (parameter sharing) fixes. The $\sim$41,000$\times$ total reduction comes from locality and sharing together.

Exercise 2.1: Dense Layer Weights for a 64×64 Image

Excluding biases, a dense layer needs $n_{\text{in}} \times n_{\text{out}}$ weights, and flattening gives $n_{\text{in}} = H \cdot W \cdot C$:

- **Grey-scale:** $64 \cdot 64 \cdot 512 = 4096 \cdot 512 = 2{,}097{,}152$ weights ($\approx 2.1 \times 10^6$).
- **RGB:** $64 \cdot 64 \cdot 3 \cdot 512 = 12{,}288 \cdot 512 = 6{,}291{,}456$ weights ($\approx 6.3 \times 10^6$).

The RGB count is exactly 3$\times$ the grey-scale count, since the input has three times as many numbers. If biases were included, you would add 512 to each, but the exercise says to exclude them.

Exercise 2.2: Three Wirings for a $224 \times 224 \times 3$ Input

The output is 64 maps of size 224×224, and a 3×3 kernel on 3 input channels sees $3 \cdot 3 \cdot 3 = 27$ inputs. Excluding biases:

- **Dense:** $(224 \cdot 224 \cdot 3) \cdot (64 \cdot 224 \cdot 224) = 150{,}528 \cdot 3{,}211{,}264 = 483{,}385{,}147{,}392 \approx 4.83 \times 10^{11}$.
- **Local, not shared:** $27 \cdot 3{,}211{,}264 = 86{,}704{,}128 \approx 8.67 \times 10^7$.
- **Local and shared:** $27 \cdot 64 = 1{,}728$.

Locality alone cuts the weights by a factor of $150{,}528/27 \approx 5{,}575.1$, and sharing then divides by the number of positions, $224 \cdot 224 = 50{,}176$. That gives a total reduction of about 279.7 million times (483,385,147,392/1,728).

If you include one bias per filter, the shared count becomes $64 \cdot (27 + 1) = 1{,}792$. The exercise says to exclude biases, so 1,728 is the answer here.

Local, Not Shared vs. Local and Shared

Both versions use the same wiring: every output unit looks at the same local $3 \times 3 \times 3$ patch (27 numbers). The only difference is whether different output units get their own weights or reuse the same ones.

The output has $64 \times 224 \times 224 = 3{,}211{,}264$ units. Each one computes a dot product over its 27 inputs.

Local, not shared: $27 \times 3{,}211{,}264$. Every output unit has its own private set of 27 weights, so the unit at position $(5, 7)$ in map 1 and the unit at $(100, 40)$ in map 1 have different weights. You count 27 weights for each of the 3,211,264 units, which gives 86,704,128. This is a thought experiment (a "locally connected" layer), not a CNN.

Local and shared: 27×64. All $224 \times 224 = 50{,}176$ units in the same output map use the same 27 weights, because that map is produced by one filter sliding over the image. Only the 64 maps differ from each other, since each map has its own filter. So you count 27 weights per filter for 64 filters, which gives 1,728. The number of positions drops out entirely.

How they relate: $86{,}704{,}128/1{,}728 = 50{,}176$, which is exactly the number of positions per map. Sharing divides the weight count by the number of spatial positions.

Sharing saves memory, not computation. In both cases the layer still performs 3,211,264 dot products of length 27, which is 86,704,128 multiply–accumulates. That number happens to equal the not-shared weight count, because there each weight is used exactly once. In the shared case each weight is used 50,176 times. This matches the point in Section 7.1 that parameters are independent of image size but MACs grow with it.

What sharing means for the network: because one filter is applied everywhere, a pattern learned at one position is detected at every position (this is Problem 3, translation). Without sharing, the unit at each position would have to learn the same edge detector separately.

Meaning of "Each Output" in Example 2.1

In Example 2.1, "each output" means **one number in the output volume**, which is one neuron.

Where 16,384 comes from. The layer produces 16 feature maps, each 32×32. Every entry in those maps is one neuron:

$$32 \cdot 32 \cdot 16 = 16{,}384 \text{ neurons.}$$

What each neuron does. Take the neuron at position (i, j) in map o. It looks at the 5×5 patch around (i, j) in the input, across all 3 colour channels. That is $5 \cdot 5 \cdot 3 = 75$ input numbers, and it computes one dot product with 75 weights:

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_{75} x_{75} \; (+ b).$$

What "not shared" means. Every one of the 16,384 neurons owns its own private 75 weights:

- The neuron at $(0, 0)$ in map 1 has weight set A.
- The neuron at $(0, 1)$ in map 1 has a different set B, even though it is in the same map and only one step over.

- The neuron at $(0, 0)$ in map 2 has yet another set C.

Nothing is reused, so the total is $16{,}384 \times 75 = 1{,}228{,}800$ weights.

Comparison with the other two cases in the same example:

- **Dense:** each of the 16,384 neurons looks at all 3,072 inputs, so $16{,}384 \times 3{,}072$ weights.
- **Local, not shared:** each neuron looks at only 75 inputs, so $16{,}384 \times 75$. Locality shrinks how many inputs each neuron sees.
- **Local and shared (CNN):** all 1,024 neurons in the same map use one set of 75 weights. Only the 16 maps have different weights, so $16 \times (75 + 1) = 1{,}216$. Sharing shrinks how many distinct weight sets exist.

So in the "not shared" case, the wiring already looks like a CNN (small patches), but the weights are still private per neuron, like an MLP.

Exercise 2.3: Why Parameter Sharing Is Poor for Aligned Passport Photos

In aligned passport photos the eyes, nose and mouth always sit in roughly the same pixel positions, so translation is not a problem. Parameter sharing becomes a poor fit for a related reason: **in aligned photos, position itself carries meaning, and sharing throws that information away.**

What sharing assumes. Sharing builds in stationarity (the Key Idea box in Section 2.3): the same pattern can appear anywhere, so the same detector should be used everywhere. The image statistics are assumed to be the same at every location.

Why that assumption breaks for passport photos.

1. **Different regions need different detectors.** The top-middle region always contains eyes, the centre a nose, the bottom a mouth. The best features for the eye region (iris shape, eyelid edges) are different from the best features for the mouth region (lip contours, teeth). With sharing, one filter must serve every position, so it cannot specialise to "what belongs here".
2. **Sharing wastes capacity.** Every filter slides over the whole image, so an eye detector also runs over the chin, the hair and the background, where its output is useless. Without sharing, each position's weights can learn only what matters at that position.
3. **The benefit of sharing disappears.** Sharing earns its place by giving translation equivariance: a pattern learned in one place is recognised in every other place. When faces never move, that benefit is unused, but the restriction (identical weights everywhere) is still paid.

What still helps. Locality is still a good assumption, since eyes and mouths are still local structures. So a reasonable design is the "local, not shared" layer from Example 2.1: small patches, but private weights per position. This is a real technique called a *locally connected layer*. For example, Facebook's DeepFace (2014) used locally connected layers after aligning faces, for exactly this reason. In practice, early layers often still share weights (edges look the same everywhere), and the unshared layers come later, where features become position-specific.

Short answer: sharing assumes the same feature means the same thing at every location. In aligned passport photos, each location has its own fixed role (eyes here, mouth there), so position-specific weights fit better, and the equivariance that sharing provides is never needed.

Exercise 2.4: Distance Between Vertical Neighbours After Flattening

Setup. A 32×32 grey image has 1024 pixels. Flattening turns it into a vector of length 1024. The question asks how far apart pixels (i, j) and $(i+1, j)$ land, which are vertical neighbours (same column, adjacent rows).

Row-major flattening (rows stacked one after another). Row 0 fills positions 0–31, row 1 fills

32–63, and so on. So pixel (i, j) goes to

$$\text{index} = 32i + j.$$

- $(i, j) \to 32i + j$
- $(i + 1, j) \to 32(i + 1) + j = 32i + j + 32$

Distance $= 32$**.** Moving down one row skips a whole row of 32 pixels.

Column-major flattening (columns stacked one after another). Column 0 fills positions 0–31, column 1 fills 32–63, and so on. So pixel (i, j) goes to

$$\text{index} = 32j + i.$$

- $(i, j) \to 32j + i$
- $(i + 1, j) \to 32j + i + 1$

Distance $= 1$**.** Vertical neighbours are in the same column, so they stay adjacent.

The point of the exercise. Horizontal neighbours (i, j) and $(i, j + 1)$ behave the opposite way: distance 1 in row-major, 32 in column-major. Whichever order you choose, one kind of neighbour gets separated by 32. A 2-D grid cannot be laid out in 1-D while keeping both directions adjacent, so flattening always destroys part of the spatial layout. An MLP sees only the vector, so it has no way to know which pixels were neighbours. A CNN avoids this by keeping the $H \times W$ layout and sliding its kernel over both axes.

3.1 One-Dimensional Cross-Correlation and Convolution

Setup. We have a signal $\mathbf{x} = (x_0, \ldots, x_{n-1})$ and a short kernel (filter) $\mathbf{k} = (k_0, \ldots, k_{m-1})$, with $m < n$. The kernel slides along the signal. At each position, the aligned entries are multiplied and the products are added. That sum is one output value.

Cross-correlation ($\star$)

$$(x \star k)_i = \sum_{u=0}^{m-1} x_{i+u}\, k_u, \qquad i = 0, \ldots, n - m$$

The kernel is laid over the signal **exactly as written**: k_0 sits on x_i, k_1 on x_{i+1}, and so on.

Convolution ($*$)

$$(x * k)_i = \sum_{u=0}^{m-1} x_{i+u}\, k_{m-1-u}$$

This is the same sliding and summing, but the kernel is **reversed first**. The last kernel element sits on x_i and the first sits on x_{i+m-1}.

Output length (valid mode). The kernel must fit entirely inside the signal, so there are $n - m + 1$ outputs.

Note on the textbook form. Signal processing usually writes convolution as $\sum_u x_{t-u} k_u$. The minus sign means "walk backwards through the input", which is the same thing as reading the kernel backwards. Formula (4) in the notes is that same quantity, restricted to fully overlapping positions and re-indexed to start at 0.

Worked example (Example 3.2)

Let $x = (1, 3, 2, 0, -1, 2)$ and $k = (1, 2, -1)$. That gives $6 - 3 + 1 = 4$ outputs.

- **Correlation**, using k as written $(1, 2, -1)$:
  - $y_0 = 1 \cdot 1 + 3 \cdot 2 + 2 \cdot (-1) = 5$
  - the full result is $x \star k = (5, 7, 3, -4)$
- **Convolution**, using the flipped kernel $(-1, 2, 1)$:
  - $y_0 = 1 \cdot (-1) + 3 \cdot 2 + 2 \cdot 1 = 7$
  - the full result is $x * k = (7, 1, -3, 0)$

Full convolution

In **valid** mode, the kernel is only placed where it fits **completely inside** the signal, which gives $n - m + 1 = 4$ positions.

In **full** mode, the kernel is placed at **every position where it overlaps the signal by at least one element**. This starts where only the kernel's last element touches x_0 and ends where only its first element touches x_{n-1}. Wherever the kernel hangs off the edge, the missing signal values are treated as 0. In practice, this means padding the signal with $m - 1$ zeros on each side.

Length. Padding gives $n + 2(m - 1) = 6 + 4 = 10$ elements. Sliding a length-3 kernel over 10 elements gives $10 - 3 + 1 = 8 = n + m - 1$ outputs.

How the result is calculated.

- Padded signal: $(0, 0, 1, 3, 2, 0, -1, 2, 0, 0)$
- Kernel $k = (1, 2, -1)$, so the flipped kernel is $(-1, 2, 1)$

Slide the flipped kernel over the padded signal, multiply the aligned entries, and add:

t	window	$\times (-1, 2, 1)$	result
0	$(0, 0, 1)$	$0 + 0 + 1$	1
1	$(0, 1, 3)$	$0 + 2 + 3$	5
2	$(1, 3, 2)$	$-1 + 6 + 2$	7
3	$(3, 2, 0)$	$-3 + 4 + 0$	1
4	$(2, 0, -1)$	$-2 + 0 - 1$	-3
5	$(0, -1, 2)$	$0 - 2 + 2$	0
6	$(-1, 2, 0)$	$1 + 4 + 0$	5
7	$(2, 0, 0)$	$-2 + 0 + 0$	-2

Full result: $(1, 5, 7, 1, -3, 0, 5, -2)$.

Why the middle 4 entries are the valid result.

- $t = 0, 1$ use left-side padding zeros, so the kernel is only partly on the signal.
- $t = 6, 7$ use right-side padding zeros, for the same reason.
- $t = 2$ to 5 have the window entirely inside the real signal. These are exactly the valid outputs $(7, 1, -3, 0)$.

This matches the identity in the notes: valid index i equals full index $i + m - 1 = i + 2$. So valid y_0 is full y_2.

Cross-check with the formula. The signal-processing form $(x * k)_t = \sum_u x_{t-u} k_u$ gives the same numbers directly. For example, at $t = 1$: $x_1 k_0 + x_0 k_1 = 3 \cdot 1 + 1 \cdot 2 = 5$. Any x index outside 0 to 5 counts as 0.

Difference between cross-correlation and convolution

	Cross-correlation	**Convolution**
Kernel orientation	As written	Flipped (in 2-D: rotated 180°)
Intuition	Template matching: "how much does this patch look like k?"	Mathematical convolution (signal processing, system response)

- **They differ only when the kernel is asymmetric.** If $k_u = k_{m-1-u}$ (for example $(1, 2, 1)$), flipping changes nothing and the two operations give identical results.
- **Practical point for CNNs.** PyTorch and TensorFlow `conv2d` actually compute **cross-correlation**. Because kernels are learned, this does not matter: the network simply learns the flipped version of whatever kernel it needs. The flip only matters when you plug in a hand-designed kernel (such as Sobel), check a library's output against a hand calculation, or derive the backward pass.

Relationship between the two operations

The relationship can be written in two directions:

$$x * k = x \star \text{flip}(k), \qquad x \star k = x * \text{flip}(k)$$

These are not two separate facts. They are two directions of the same fact. Each one says how to compute one operation using the other.

Why both hold. Convolution is defined as correlation with a flipped kernel, so $x * k = x \star \text{flip}(k)$. Flipping twice restores the original: $\text{flip}(\text{flip}(k)) = k$. Substituting $\text{flip}(k)$ for k:

$$x * \text{flip}(k) = x \star \text{flip}(\text{flip}(k)) = x \star k$$

That gives the other direction. One identity implies the other because flip is its own inverse.

Check with the example. $k = (1, 2, -1)$, so $\text{flip}(k) = (-1, 2, 1)$.

- $x \star \text{flip}(k)$ uses kernel $(-1, 2, 1)$ as written, which gives $(7, 1, -3, 0)$. That equals $x * k$.
- $x * \text{flip}(k)$ flips $(-1, 2, 1)$ back to $(1, 2, -1)$, which gives $(5, 7, 3, -4)$. That equals $x \star k$.

Theorem 3.4 states this same relationship in 2-D, using rot180.

3.2 Two-Dimensional Cross-Correlation and Convolution

This is the same idea as 1-D, except the kernel is now a small matrix K of size $k_h \times k_w$. It slides across the image both horizontally and vertically.

- **Cross-correlation:**
$$(X \star K)_{i,j} = \sum_u \sum_v x_{i+u,\, j+v}\, k_{u,v}$$
  - Place K on the patch whose top-left corner is (i, j).
  - Multiply the aligned entries and add them up.
- **Convolution:** the same operation, but K is flipped along **both** axes first. A flip along both axes is a 180° **rotation**:
$$K = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \qquad \text{rot180}(K) = \begin{bmatrix} i & h & g \\ f & e & d \\ c & b & a \end{bmatrix}$$
- **Output size (valid):** $(H - k_h + 1) \times (W - k_w + 1)$

- **Theorem 3.4:** $X * K = X \star \text{rot180}(K)$, and the reverse also holds. This is the 2-D version of the flip relationship from 3.1.

Concrete example (Example 3.8).

$$X = \begin{bmatrix} 2 & 0 & 1 & 3 \\ 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}, \qquad K = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

The output is $(4 - 2 + 1) \times (4 - 2 + 1) = 3 \times 3$.

- **Correlation at** $(0, 0)$**:** the patch is $\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$.
  - $2 \cdot 1 + 0 \cdot 2 + 1 \cdot 0 + 3 \cdot (-1) = -1$
- **Convolution at** $(0, 0)$**:** use $\text{rot180}(K) = \begin{bmatrix} -1 & 0 \\ 2 & 1 \end{bmatrix}$ on the same patch.
  - $2 \cdot (-1) + 0 \cdot 0 + 1 \cdot 2 + 3 \cdot 1 = 3$

Doing this at all 9 positions gives:

$$X \star K = \begin{bmatrix} -1 & 2 & 5 \\ 6 & 1 & 3 \\ 0 & 4 & 4 \end{bmatrix}, \qquad X * K = \begin{bmatrix} 3 & 6 & 1 \\ 0 & 1 & 5 \\ 8 & 4 & 0 \end{bmatrix}$$

The two outputs are completely different, because K is asymmetric.

3.3 What Deep-Learning Libraries Compute

`conv2d` in PyTorch and TensorFlow computes **cross-correlation**, even though the layer is called "convolutional".

This is harmless because the kernel is **learned**. Suppose the ideal filter under true convolution is K'. A network using correlation simply learns $\text{rot180}(K')$ instead, and the outputs are identical.

The flip matters in only three situations:

1. You copy a hand-designed filter (Sobel, for example) from a signal-processing textbook.
2. You check library output against a hand calculation.
3. You derive backpropagation, where both operations appear.

From this point on, "convolution" in a CNN means cross-correlation.

3.4 Properties of the Operation

Proposition 3.5: Basic properties

All four examples below use small 1-D numbers so the arithmetic is easy to follow. Each property holds the same way in 2-D.

(1) Linear in the input

Statement: $(\alpha X + \beta X') \star K = \alpha(X \star K) + \beta(X' \star K)$

In plain words: mixing two inputs and then filtering gives the same result as filtering each input and then mixing the outputs.

Example: $x = (1, 2, 3)$, $x' = (0, 1, 0)$, $k = (1, -1)$. With $k = (1, -1)$, each output is $x_i - x_{i+1}$.

- $x \star k = (1 - 2,\ 2 - 3) = (-1, -1)$

- $x' \star k = (0-1,\ 1-0) = (-1, 1)$
- **Mix first, then filter:**
  - $2x + x' = (2, 5, 6)$
  - $(2x + x') \star k = (2-5,\ 5-6) = (-3, -1)$
- **Filter first, then mix:**
  - $2(-1, -1) + (-1, 1) = (-3, -1)$, which matches.

Why it holds: every term in the sum is one input value times one kernel value. Scaling or adding inputs just passes straight through.

(2) Linear in the kernel

Statement: $X \star (\alpha K + \beta K') = \alpha(X \star K) + \beta(X \star K')$

This is the same idea with the roles of input and kernel swapped.

Example: $x = (1, 2, 3)$, $k = (1, -1)$, $k' = (1, 1)$.

- $x \star k = (-1, -1)$
- $x \star k' = (3, 5)$
- Adding the outputs: $(-1, -1) + (3, 5) = (2, 4)$
- Adding the kernels first: $k + k' = (2, 0)$
  - $x \star (2, 0) = (2 \cdot 1,\ 2 \cdot 2) = (2, 4)$, which matches.

What "the output is linear in the weights" means. Treat the input X as fixed and the kernel entries $k_{u,v}$ as the variables. Each output is then

$$y_{ij} = \sum x_{i+u,\, j+v}\, k_{u,v}$$

This is a plain weighted sum of the weights, where the input pixels act as fixed coefficients. There are no squares, products, or other nonlinear functions of the weights.

What "linear" guarantees:

- **Scaling:** doubling the kernel doubles the output, since $X \star (2K) = 2(X \star K)$.
- **Adding:** using kernel $K + K'$ gives the same result as running both kernels separately and adding the outputs.

Concrete example: take the patch $(3, 5)$ with kernel (k_0, k_1). The output is $y = 3k_0 + 5k_1$.

- With kernel $(1, 0)$, $y = 3$.
- With kernel $(0, 1)$, $y = 5$.
- With kernel $(2, 1)$, $y = 2 \cdot 3 + 1 \cdot 5 = 11$.

The output responds to each weight in a straight-line way.

Why it matters for training: the derivative is trivial.

- $\partial y_{ij} / \partial k_{u,v} = x_{i+u,\, j+v}$. The gradient with respect to a weight is just the input pixel it multiplies.
- In the example above, $\partial y / \partial k_0 = 3$ and $\partial y / \partial k_1 = 5$.

This is why the backward pass gives $\partial L / \partial K = X \star \delta$, a correlation of the input with the incoming gradient. The whole network is not linear in its weights, because layers and activations stack. Each single conv operation, though, is linear in its own weights.

(3) Commutativity: true for convolution, not for correlation

Statement: $X * K = K * X$ holds for full convolution. It does not hold for correlation.

Example: $x = (1, 2)$, $k = (3, 4)$, in full mode.

- **Convolution:**
  - $x * k = (1 \cdot 3,\ 1 \cdot 4 + 2 \cdot 3,\ 2 \cdot 4) = (3, 10, 8)$
  - $k * x$ gives $(3, 10, 8)$ as well. Swapping the arguments changes nothing.
  - A useful way to see this: full convolution is polynomial multiplication. $(1 + 2z)(3 + 4z) = 3 + 10z + 8z^2$, and polynomial multiplication does not care about order.
- **Correlation** (full mode, kernel slid as written):
  - $x \star k = (4, 11, 6)$
  - $k \star x = (6, 11, 4)$
  - The results are **reversed**, not equal. Swapping the roles mirrors the output, which is exactly what the proof says ("evaluated at $-i$").

(4) Associativity: why CNNs need nonlinearities

Statement: $(X * K) * K' = X * (K * K')$

In plain words: applying two filters one after the other is the same as applying a single combined filter.

Example: $x = (1, 2, 4)$, $k = (1, 1)$, $k' = (1, -1)$.

- **Two steps:**
  - $x * k = (1, 3, 6, 4)$
  - $(1, 3, 6, 4) * k' = (1, 2, 3, -2, -4)$
- **One combined kernel:**
  - $k * k' = (1, 0, -1)$
  - $x * (1, 0, -1) = (1, 2, 3, -2, -4)$, which matches.

The key consequence: two stacked conv layers with **no activation between them** collapse into one conv layer with a bigger kernel. Here, sizes 2 and 2 combine into size $2 + 2 - 1 = 3$. The extra layer adds no expressive power. A ReLU between the layers breaks this collapse, which is why every conv layer is followed by a nonlinearity.

Does associativity hold for valid convolution?

Yes, with the right combined kernel. Stacking two valid convolutions equals one valid convolution with the combined kernel $k * k'$. That combined kernel must be computed with **full** convolution, so its size is $m + m' - 1$.

$$(x *_{\text{valid}} k) *_{\text{valid}} k' = x *_{\text{valid}} (k *_{\text{full}} k')$$

The sizes match: $n \to n - m + 1 \to n - m - m' + 2$. Applying one valid convolution with a kernel of size $m + m' - 1$ gives $n - (m + m' - 1) + 1$, which is the same.

Check: $x = (1, 2, 4)$, $k = (1, 1)$, $k' = (1, -1)$.

- **Two steps:**
  - $x *_{\text{valid}} k$ uses the flipped kernel $(1, 1)$, giving $(3, 6)$.
  - $(3, 6) *_{\text{valid}} k'$ uses the flipped kernel $(-1, 1)$, giving $-3 + 6 = 3$.
- **One step:**
  - $k * k' = (1, 0, -1)$, and its flip is $(-1, 0, 1)$.
  - $x *_{\text{valid}} (1, 0, -1) = -1 + 0 + 4 = 3$, which matches.

The valid outputs are the middle entries of the full result $(1, 2, 3, -2, -4)$ computed above.

Does associativity hold for cross-correlation?

The literal form $(x \star k) \star k' = x \star (k \star k')$ is false in general. Using the same numbers:

- **Left side:**
  - $x \star k = (3, 6)$
  - $(3, 6) \star (1, -1) = 3 - 6 = -3$
- **Right side:**
  - $k \star k'$ in full mode $= (-1, 0, 1)$
  - $x \star (-1, 0, 1) = -1 + 4 = 3$, which does not match.

The two sides have opposite signs.

The correct version: $(x \star k) \star k' = x \star (k * k')$. The combined kernel uses true convolution of the two kernels.

- **Check:** $x \star (1, 0, -1) = 1 - 4 = -3$, which matches.
- **Why it holds:**
  - Correlation is convolution with a flipped kernel.
  - $\text{flip}(k) * \text{flip}(k') = \text{flip}(k * k')$.
  - So the two flips merge into one flip of the combined kernel.

The practical conclusion is unchanged. Two stacked correlation layers (as used in PyTorch and TensorFlow) without an activation between them always reduce to one correlation layer with a larger kernel. That is why a nonlinearity is required between layers.

Theorem 3.6: Translation equivariance

Statement: $(TX) \star K = T(X \star K)$, where T shifts the image.

In plain words: if you shift the input and then filter it, you get the same result as filtering first and then shifting the output by the same amount. **If the object moves, its response moves with it, unchanged.**

Example (1-D, $k = (1, -1)$, so each output is $x_i - x_{i+1}$):

- Original: $x = (0, 0, 5, 1, 0, 0, 0)$
  - $x \star k = (0, -5, 4, 1, 0, 0)$
- Shift x right by 2: $(0, 0, 0, 0, 5, 1, 0)$
  - $\star k = (0, 0, 0, -5, 4, 1)$

The pattern $(-5, 4, 1)$ is identical in both results. It has just moved right by 2.

Why it holds: the **same kernel** is used at every position (parameter sharing). So the filter cannot tell *where* a pattern sits. It only reports *what* it found, and the location of that report follows the pattern.

Two important contrasts:

- **Dense layer:** each pixel has its own weights. A shifted cat hits completely different weights, so the output can change arbitrarily. A dense layer has no equivariance.
- **Equivariance vs invariance:**
  - Equivariant means the output **moves** with the input.

– Invariant means the output **does not change** at all.

– Convolution is equivariant only. Approximate invariance to small shifts comes later from pooling (Section 9).

3.5 A First Manual Example

This section is the first hand calculation of a 2-D cross-correlation. It shows three things: the mechanics, what a kernel "detects", and how correlation and convolution can differ.

Example 3.7.

$$X = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}, \qquad K = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The setup uses stride 1 and no padding.

Step 1, output size: the kernel fits at $3 - 2 + 1 = 2$ positions along each axis, so the output Y is 2×2.

Step 2, compute each output: place K on each 2×2 patch, multiply the aligned entries, and add.

- K has 1 at the top-left, -1 at the bottom-right, and 0 elsewhere.
- So each output is simply **top-left − bottom-right**.

position	patch	calculation	y
$(0,0)$	$\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$	$1 - 0$	1
$(0,1)$	$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$	$2 - 1$	1
$(1,0)$	$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$	$1 - 1$	0
$(1,1)$	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$	$0 - 1$	-1

$$Y = X \star K = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

Step 3, interpret the result: $y_{ij} = x_{ij} - x_{i+1,j+1}$ is a **diagonal difference**.

- A patch with the same brightness along the diagonal gives 0.
- A patch that gets darker toward the bottom-right gives a positive output.
- A patch that gets brighter toward the bottom-right gives a negative output.

So the kernel is a detector for edges across the main diagonal.

Step 4, compare with true convolution: $\text{rot180}(K) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -K$.

- By Theorem 3.4, $X * K = X \star (-K)$.
- By linearity, $X \star (-K) = -Y = \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix}$.
- The magnitudes are the same but every sign is flipped. With a library correlation versus a textbook convolution, you would see the opposite sign on the edge response.

Example 3.8 (the 4×4 case shown in 3.2) is the general situation. Its kernel is not $\pm$ its own rotation, so the two outputs are completely different, not just sign-flipped.

3.6 Why Use This Operator at All?

1. Local statistics

The patterns that matter at the lowest level (edges, corners, textures) are determined by a pixel and its **immediate neighbours**.

- **Example:** a vertical edge at pixel $(100, 50)$ depends on pixels $(100, 49)$ and $(100, 51)$, not on a pixel in the opposite corner.
- So a 3×3 kernel that sees only a small neighbourhood has all the information it needs.
- Connecting each output to every pixel, as a dense layer does, adds weights that carry mostly irrelevant information.

2. Stationarity

The **same pattern means the same thing anywhere** in the image. An edge is an edge whether it appears top-left or bottom-right.

- So one detector should be reused at every location, which is exactly what sliding a single kernel does.
- **Example:** a Sobel kernel slid over a whole photo finds every vertical edge with just 9 weights.
- A dense layer would need to learn "edge at location A" and "edge at location B" separately.
- This is also what gives the translation equivariance of Theorem 3.6.
- The assumption can fail. In aligned passport photos the eyes are always in the same place, so location-specific weights could help (see Exercise 2.3).

3. Composition

Each layer sees a small window, but stacking layers **widens the receptive field** (Section 10).

- **Example:** two 3×3 layers mean each output depends on a 5×5 region of the input. Three layers give 7×7.
- With nonlinearities between them, the layers build a hierarchy:
  - Layer 1 detects edges.
  - Layer 2 combines edges into corners and curves.
  - Layer 3 combines those into parts such as an eye or a wheel.
  - Deeper layers detect whole objects.
- Without the nonlinearities, associativity collapses the stack into one big linear filter and the hierarchy disappears. This connects directly to Proposition 3.5 (4).

4. Efficiency

Local connections and shared weights give **far fewer parameters** (Section 7).

- **Example:** take a 224×224 grey image mapped to a 224×224 output.
  - **Dense layer:** $224^2 \times 224^2 \approx 2.5$ billion weights.
  - **One 3×3 conv kernel:** 9 weights plus 1 bias, regardless of image size.
- Fewer parameters mean less memory, faster training, and less overfitting.

Key idea: a conv layer is a bank of linear filters with tied (shared) weights. The nonlinearity after each layer is what turns that bank into a hierarchical pattern detector. Without the nonlinearity,

associativity would collapse the whole stack into one filter.

3.7 Summary

- Cross-correlation: $(X \star K)_{ij} = \sum x_{i+u,\, j+v}\, k_{u,v}$. True convolution is the same with K rotated 180°: $X * K = X \star \mathrm{rot180}(K)$.
- Valid output size: $(H - k_h + 1) \times (W - k_w + 1)$.
- Libraries compute cross-correlation. Because kernels are learned, the flip does not matter for training.
- The operation is linear in both the input and the kernel, and it is translation equivariant.

Solutions to Exercises 3.1 to 3.5: Detailed Breakdown

These are the five exercises at the end of Section 3 (The Convolution Operation) of the Week 3 lecture notes.

Exercise 3.1: Valid Correlation and Convolution in 1-D

Given: $x = (2, -1, 0, 3, 1)$, so $n = 5$. $k = (1, 0, 2)$, so $m = 3$.

Step 1: How many outputs? A valid output exists only where the whole kernel fits inside the signal. That gives $n - m + 1 = 5 - 3 + 1 = 3$ outputs ($i = 0, 1, 2$).

Step 2: Cross-correlation, with the kernel as written $(1, 0, 2)$**.** The formula is $y_i = x_i k_0 + x_{i+1} k_1 + x_{i+2} k_2$.

i	window of x	computation	y_i
0	$(2, -1, 0)$	$2 \cdot 1 + (-1) \cdot 0 + 0 \cdot 2$	**2**
1	$(-1, 0, 3)$	$(-1) \cdot 1 + 0 \cdot 0 + 3 \cdot 2$	**5**
2	$(0, 3, 1)$	$0 \cdot 1 + 3 \cdot 0 + 1 \cdot 2$	**2**

So $x \star k = (2, 5, 2)$.

Step 3: Convolution, which flips the kernel first. Reversing $k = (1, 0, 2)$ gives $(2, 0, 1)$. Then the flipped kernel slides exactly as in correlation.

i	window of x	computation	y_i
0	$(2, -1, 0)$	$2 \cdot 2 + (-1) \cdot 0 + 0 \cdot 1$	**4**
1	$(-1, 0, 3)$	$(-1) \cdot 2 + 0 \cdot 0 + 3 \cdot 1$	**1**
2	$(0, 3, 1)$	$0 \cdot 2 + 3 \cdot 0 + 1 \cdot 1$	**1**

So $x * k = (4, 1, 1)$.

Why the two results differ: k is not symmetric, because $(1, 0, 2)$ reversed is $(2, 0, 1)$, which is a different kernel. Correlation and convolution give the same result only when the kernel reads the same in both directions.

Exercise 3.2: When Correlation Equals Convolution

Part 1: The proof.

- Theorem 3.4 says $X * K = X \star \mathrm{rot180}(K)$. In words, convolution is correlation with the kernel rotated by 180°.
- Suppose $K = \mathrm{rot180}(K)$. Substituting that into the theorem gives $X * K = X \star K$. ■

Convolution differs from correlation only by the rotation, so a kernel that the rotation leaves unchanged makes the two operations identical.

Part 2: What the condition means entry by entry. For a 3×3 kernel, rot180 maps position (u, v) to $(2-u, 2-v)$. So the condition is $k_{u,v} = k_{2-u,\,2-v}$, which means each entry must equal the entry diagonally opposite it through the centre:

- the top-left corner equals the bottom-right corner,
- the top-right corner equals the bottom-left corner,
- the top-middle entry equals the bottom-middle entry,
- the left-middle entry equals the right-middle entry,
- the centre entry is free.

This property is called **point symmetry**. It is weaker than mirror symmetry about an axis.

Part 3: An example that is not symmetric about either axis.

$$K = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 4 & 3 \\ 0 & 2 & 1 \end{pmatrix}$$

- **rot180**(K)**:** Reverse the row order, then reverse each row. The last row $(0, 2, 1)$ becomes $(1, 2, 0)$, the middle row $(3, 4, 3)$ stays $(3, 4, 3)$, and the first row $(1, 2, 0)$ becomes $(0, 2, 1)$. The result equals K.
- **Left-right mirror:** The first row becomes $(0, 2, 1)$, which differs from $(1, 2, 0)$. So K is not symmetric about the vertical axis.
- **Up-down mirror:** The first row becomes $(0, 2, 1)$, which again differs. So K is not symmetric about the horizontal axis.

Exercise 3.3: A 2-D Correlation and What the Kernel Detects

Given:

$$X = \begin{pmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{pmatrix}, \qquad K = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$

Step 1: Output size. $(3 - 2 + 1) \times (3 - 2 + 1) = 2 \times 2$.

Step 2: General formula for this kernel. Placing K on the 2×2 patch whose top-left corner is at (i, j) gives

$$y_{ij} = (x_{ij} + x_{i+1,j}) - (x_{i,j+1} + x_{i+1,j+1}).$$

In words, this is the sum of the left column of the patch minus the sum of the right column.

Step 3: Compute each output.

position	patch	left col − right col	y
$(0, 0)$	$\begin{pmatrix} 0 & 1 \\ 3 & 4 \end{pmatrix}$	$(0 + 3) - (1 + 4)$	$\mathbf{-2}$
$(0, 1)$	$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix}$	$(1 + 4) - (2 + 5)$	$\mathbf{-2}$
$(1, 0)$	$\begin{pmatrix} 3 & 4 \\ 6 & 7 \end{pmatrix}$	$(3 + 6) - (4 + 7)$	$\mathbf{-2}$
$(1, 1)$	$\begin{pmatrix} 4 & 5 \\ 7 & 8 \end{pmatrix}$	$(4 + 7) - (5 + 8)$	$\mathbf{-2}$

$$X \star K = \begin{pmatrix} -2 & -2 \\ -2 & -2 \end{pmatrix}$$

Step 4: Interpretation.

- The kernel measures the **horizontal change in intensity**, so it **detects vertical edges**, in the same way as Sobel-x but without the smoothing weights.
- With cross-correlation, the output is positive when brightness *decreases* from left to right and negative when it *increases*.
- In this X, every row increases by exactly 1 per column, so the image is a uniform left-to-right ramp. Each patch therefore has the same response, 2 rows $\times$ $(-1) = -2$. The output is constant because the gradient is constant.

Exercise 3.4: Associativity of Full Convolution

The definition used: Full 1-D convolution is $(x * k)_t = \sum_u x_{t-u} k_u$, summed over every u where the indices are valid. All sums are finite.

Step 1: Expand the left-hand side. Treat $(x * k)$ as the signal and convolve it with k':

$$((x * k) * k')_t = \sum_s (x * k)_{t-s}\, k'_s = \sum_s \sum_u x_{t-s-u}\, k_u\, k'_s.$$

Step 2: Change variables. Let $r = s + u$, so the total offset into x is r, and $u = r - s$:

$$= \sum_r x_{t-r} \left(\sum_s k_{r-s}\, k'_s \right).$$

Swapping the order of summation is legal here because the sums are finite.

Step 3: Recognise the inner sum. By the same definition, $\sum_s k_{r-s} k'_s = (k * k')_r$. Therefore

$$= \sum_r x_{t-r}\, (k * k')_r = (x * (k * k'))_t. \qquad \blacksquare$$

Step 4: Kernel size of the merged kernel. The full convolution of a length-m kernel with a length-m' kernel has length $m + m' - 1$. For two 3-tap kernels, the merged kernel has 5 taps.

Step 5: Conclusion about stacked layers.

- Two conv layers with no activation compute $x * k$, then $(x * k) * k'$, which equals $x * (k * k')$. That is **one** conv layer with kernel size $k + k' - 1$.
- Biases do not change this result: $(x * k + b_1) * k' + b_2 = x * (k * k') + (b_1 \sum k' + b_2)$. The extra term is just a new constant bias.
- Libraries compute correlation rather than convolution, but correlation is convolution with a flipped kernel. A composition of flipped kernels is still a single kernel, so the conclusion still holds.
- **The point of the exercise:** Without a nonlinearity between layers, extra depth adds no expressive power. The stack collapses into one linear filter. This is why ReLU or another activation sits between conv layers.

Exercise 3.5: Checking Translation Equivariance Numerically

Setup from Example 3.8:

$$X = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}, \quad X \star K = \begin{pmatrix} -1 & 2 & 5 \\ 6 & 1 & 3 \\ 0 & 4 & 4 \end{pmatrix}$$

Step 1: Shift X one column to the right. Insert a zero column on the left and drop the last column so the image stays 4×4:

$$X' = \begin{pmatrix} 0 & 2 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 3 & 2 & 1 \end{pmatrix}$$

Step 2: Formula for this kernel.

$$y_{ij} = 1 \cdot x_{ij} + 2 \cdot x_{i,j+1} + 0 \cdot x_{i+1,j} + (-1) \cdot x_{i+1,j+1}$$

Step 3: Compute the 3×3 output.

	$j = 0$	$j = 1$	$j = 2$
$i = 0$	$0 + 4 - 1 = \mathbf{3}$	$2 + 0 - 3 = \mathbf{-1}$	$0 + 2 - 0 = \mathbf{2}$
$i = 1$	$0 + 2 - 0 = \mathbf{2}$	$1 + 6 - 1 = \mathbf{6}$	$3 + 0 - 2 = \mathbf{1}$
$i = 2$	$0 + 0 - 3 = \mathbf{-3}$	$0 + 2 - 2 = \mathbf{0}$	$1 + 4 - 1 = \mathbf{4}$

$$X' \star K = \begin{pmatrix} 3 & -1 & 2 \\ 2 & 6 & 1 \\ -3 & 0 & 4 \end{pmatrix}$$

Step 4: Compare with the original output.

- Columns 1–2 of the new output, $(-1, 6, 0)$ and $(2, 1, 4)$, are exactly columns 0–1 of the original output. **The feature map moved one column to the right**, which is what Theorem 3.6 predicts: shifting and then correlating equals correlating and then shifting.
- Column 0, $(3, 2, -3)$, is new. It comes from windows that cover the inserted zeros, so it is a **border effect**, not a violation of the theorem.
- The original output's last column, $(5, 3, 4)$, disappeared because the input column that produced it was dropped.
- This is why the theorem says to "ignore borders". Equivariance holds exactly in the interior and fails only where the padding or cropping changes the input.

Takeaway: The response pattern keeps its shape and simply moves with the input. That is equivariance. It is not invariance, because the output's position changes. Pooling adds approximate invariance later.

Pixel Values and Brightness in Image Tensors

A higher value usually means a brighter pixel, but this is a convention, not a rule built into the tensor.

- **Grayscale:** In the standard convention, 0 is black and the maximum is white. The maximum is 255 for `uint8` data, or 1.0 after scaling to $[0, 1]$. So a higher value means a brighter pixel.
- **RGB:** Each channel stores the intensity of one colour. A higher value means more of that colour. Overall brightness depends on all three channels together; a common weighted formula is luminance $\approx 0.299R + 0.587G + 0.114B$. A high value in a single channel means "more red", for example, not "brighter".

Two common exceptions:

- **Normalised inputs:** Pipelines often standardise each channel, for example $(x - \text{mean})/\text{std}$ as in ImageNet preprocessing. Values then become negative or positive. Higher still means brighter, but 0 now means "average brightness", not black.
- **Inverted conventions:** Some datasets store ink as high values on a dark background. MNIST is an example: the digit strokes are near 255 and the background is 0. A higher value then means

"more ink", which looks white when displayed but represents the dark stroke in the original writing. Some medical and scientific formats also use their own mappings.

For natural photos in `uint8` or $[0, 1]$ format, a higher value means brighter. Otherwise, check the dataset's preprocessing.